

Descriptor taxonomy

IDENTICAL (definitional)

$L^1 \equiv R_a$

$L^2 \equiv R_q$

ISO SHAPE (distribution)

Rsk — asymmetry

Rku — peakedness

BANACH GRADIENT (independent)

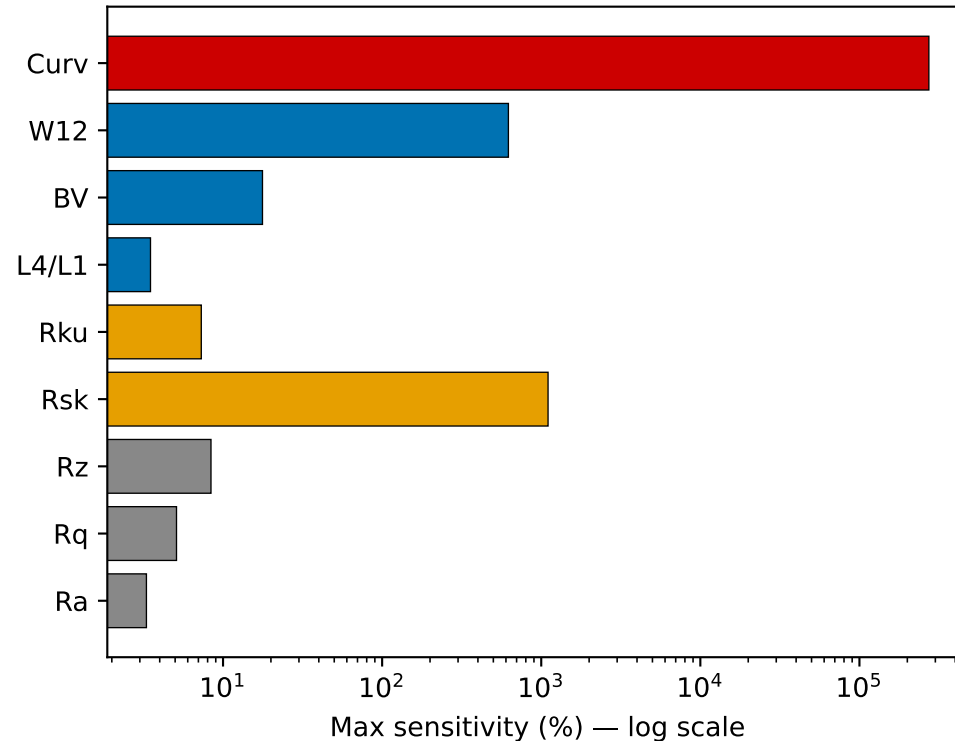
BV density — localised defects

$W^{1,2}$ grad — sharpness

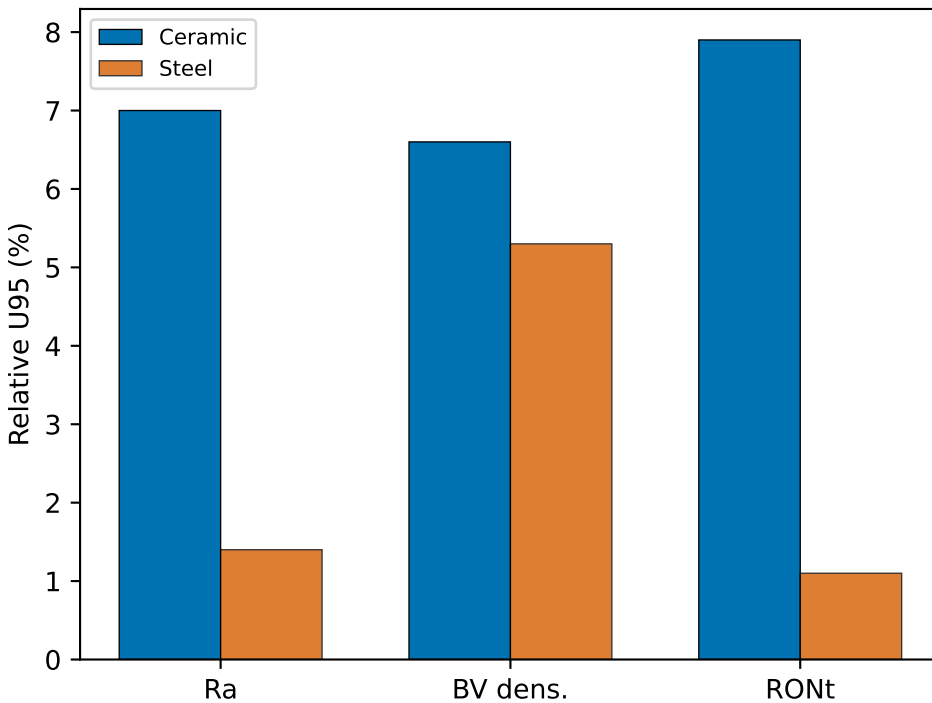
L^4/L^1 — peakiness ratio

Curvature RMS — noise-amplified

Maximum sensitivity to synthetic defects



Sweep repeatability (LSCI + 0-15 UPR)



VALIDATION SUMMARY

Ceramic En = 0.61 (n=104 tactile ref.)

Steel: 42.07 vs 41.94 μm (cert.)

→ 0.3% agreement

KEY FINDING:

BV density +17.7% for 100nm spike

Ra = 0.0% for same defect

→ Gradient descriptors see what
amplitude descriptors miss

All results: LSCI + 0-15 UPR

ISO 4291 / ISO 4287 compliant